

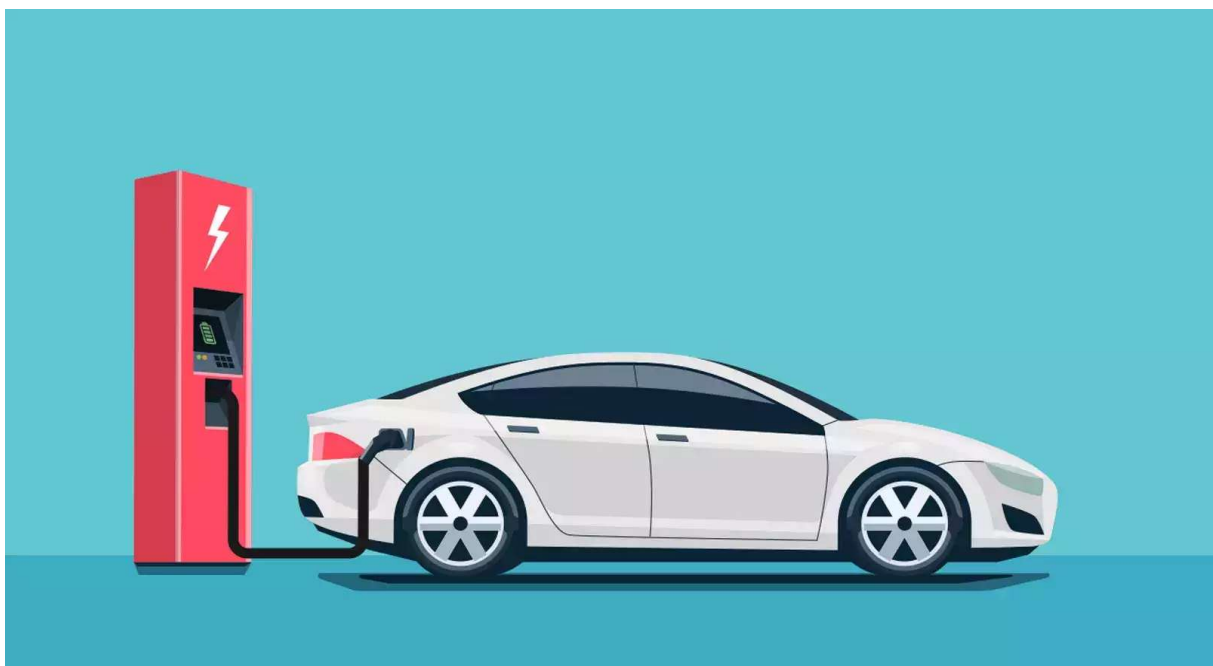
DATA ANALYSIS USING PYTHON

Assignment Name: Exploratory Data Analysis of Electric Vehicle Population in Washington State.

Course Title: Data Analytics B23

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Introduction & Objectives:

- This dataset shows the Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) that are currently registered through the Washington State Department of Licensing (DOL).
- Handled Missing values, Zero values, duplicate records and inconsistent data for data cleaning Process.
- Analysed Electric vehicle trends across Wahington state.
- Investigated regional EV adoption using Geospatial data.
- Identified EV distribution across by county, Model year, Make and Model.
- Found Clean Alternative Fuel Vehicle (CAFV) Eligible and non-Eligible distribution.
- Pricing trends by comparing Electric range and Base MSRP.
- Performed Linear Regression and found R^2 score of the model.

Section 1 -Data Cleaning:

1.Missing values exist in the dataset, and in which columns?

Column Name	Missing Values Count
1. Country	9
2. City	9
3. Postal Code	9
4. Electric Range	10
5. Base MSRP	111698
6. Legislative District	727
7. Vehicle Location	16
8. Electric Utility	9
9. 2020 Census Tract	9

2.How should missing or zero values in the Base MSRP and Electric Range columns be handled?

- Very less missing (null) values count so I replaced it with mean value for both columns.
- There are no Zero values in Electric range columns.
- But in Base MSRP column ,1837 rows are having Zero values. It will affect the statistical calculations. I replaced the **Base MSRP with median value**, because for price ranges median will be the best choice.

Example:

```
ev["Base MSRP"] = ev["Base MSRP"].fillna(ev["Base MSRP"].mean())
```

3.Are there duplicate records in the dataset? If so, how should they be managed?

Overall, there are no duplicate records in the dataset. But here the ultimate goal is EV vehicle adoption. So based on the VIN number, Make, Model, Model Year and Electric vehicle type sorted the values and dropped the duplicates and kept last updated value.

Total Duplicates-271681

Unique Records-17883

```
ev.drop_duplicates(subset=[  
    'VIN (1-10)', 'Model Year', 'Make', 'Model', 'Electric Vehicle Type',  
    'Electric Range', 'Clean Alternative Fuel Vehicle (CAFV)  
    Eligibility'], keep='last', ignore_index=True, inplace=True  
)
```

4. How can VINs be anonymized while maintaining uniqueness?

Anonymizing Vehicle Identification Numbers (VINs) requires a balance between privacy and data utility. we must hide the true identity of the vehicle while ensuring that each unique car can still be tracked over time. Here we applied **Hashing with a Salt method** to create a hidden random string so outsiders cannot guess or reverse the hash via rainbow tables.

Encode the combined salt and VIN string, then extract the hexadecimal digest.

Ex:

Test uniqueness mapping

```
original_vin = "5UX43EU07S"  
anon_vin = anonymize_vin(original_vin, salt)  
print(anon_vin)
```

Result:

e39d5ef0649696155e0ac4c3eb589945342b9490cf98b78094a272005ae6c79d

5.How can Vehicle Location (GPS coordinates) be cleaned or converted for better readability?

- Ex- POINT (-122.33 47.60)
- Replaced Vehicle Location column value string point with empty and split Longitude and Latitude and that could be imported with the new column.
- Imported Numpy and replaced null values with **dummy coordinates like 0.00**.

	longitude	latitude
0	-120.26226	46.40199
1	-122.61158	47.62691
2	-122.17743	47.41185

Section 2 - Data Exploration:

1. Top 5 most common EV makes and models in the dataset:

```
Top 5 EV Make and Model:
      Make      Model  Count
196  VOLVO      XC90    589
169  TESLA    MODEL X    564
168  TESLA    MODEL S    557
69   FORD  MUSTANG MACH-E    548
195  VOLVO      XC60    465
```

2. Distribution of EVs by county:

King county has the most registrations.

```
County
King      6102
Snohomish 2560
Pierce    2301
Clark     1040
Spokane   741
...
Morris     1
Monterey   1
Middlesex  1
Meade      1
Carroll    1
Name: count, Length: 140, dtype: int64
County with the most registrations: ('King',)
```

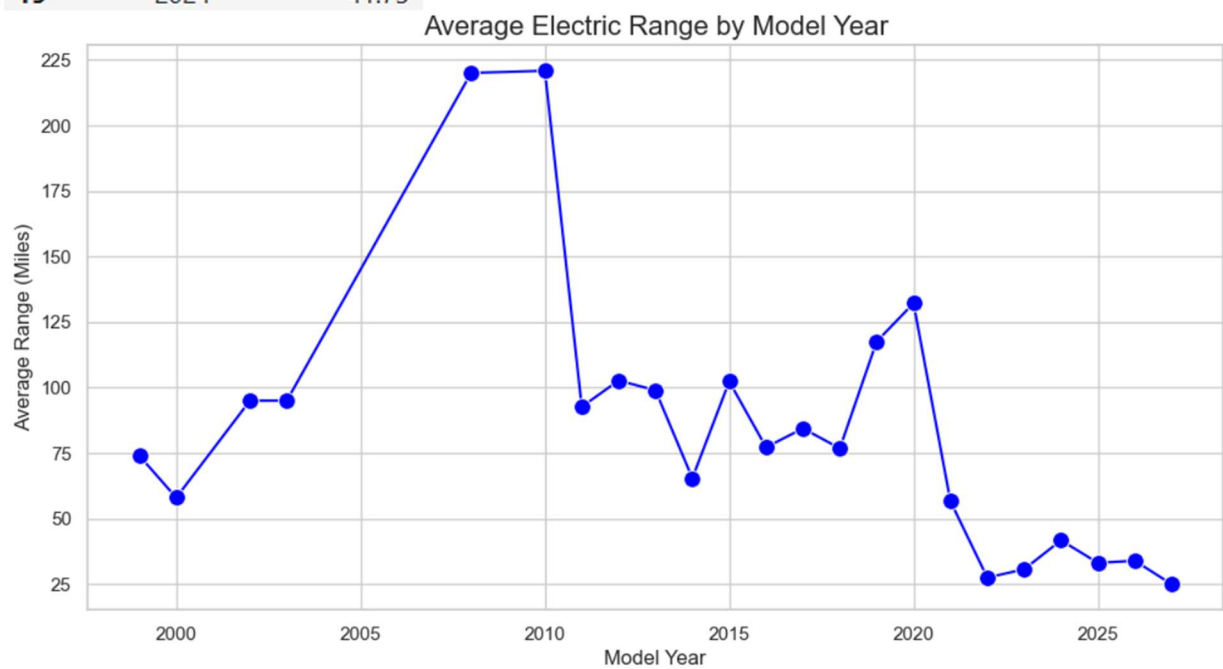
3. EV adoption changed over different model years:

EV Adoption by Model Year

Model Year	
1999	2
2000	5
2002	1
2003	1
2008	10
2010	12
2011	38
2012	153
2013	257
2014	322
2015	363
2016	463
2017	561
2018	692
2019	737
2020	821
2021	1219
2022	1866
2023	2351
2024	2956
2025	3073
2026	1935
2027	45

4.Average electric range of EVs in the dataset:

	Model Year	Electric Range
5	2010	220.83
4	2008	220.00
15	2020	132.13
14	2019	117.58
7	2012	102.69
10	2015	102.32
8	2013	98.93
2	2002	95.00
3	2003	95.00
6	2011	92.63
12	2017	84.41
11	2016	77.20
13	2018	76.80
0	1999	74.00
9	2014	65.41
1	2000	58.00
16	2021	56.64
19	2024	41.73



5. Percentage of EVs are eligible for Clean Alternative Fuel Vehicle (CAFV) incentives:

Eligibility unknown as battery range has not been researched-	9469
Clean Alternative Fuel Vehicle Eligible	- 5273
Not eligible due to low battery range	- 3141

Percentage of Eligible Ev Count=29.49%

6. Electric range vary across different makes and models:

=== AVERAGE RANGE BY MAKE & MODEL ===

Make	Model	
PORSCHE	MACAN	298.00
TESLA	MODEL Y	291.00
HYUNDAI	KONA	258.00
TESLA	MODEL 3	247.51
	MODEL X	244.96
CHEVROLET	BOLT EV	244.95
TESLA	ROADSTER	236.35
JAGUAR	I-PACE	234.00
POLESTAR	POLESTAR 2	233.00
TESLA	MODEL S	226.37

7. Average Base MSRP for each EV model:

]:

	Model	Base MSRP
0	296	52900.00
1	330E	69707.35
2	500	54950.00
3	530E	69900.00
4	550E	69900.00
5	740E	69900.00
6	745E	69900.00
7	745LE	69900.00
8	750E	69900.00
9	918	69900.00
10	A3	57000.00
11	A7 E	57000.00
12	A8 E	57000.00
13	ACCORD	59900.00
14	AMG GT	59900.00

8. Regional trends in EV adoption (e.g., urban vs. rural areas)

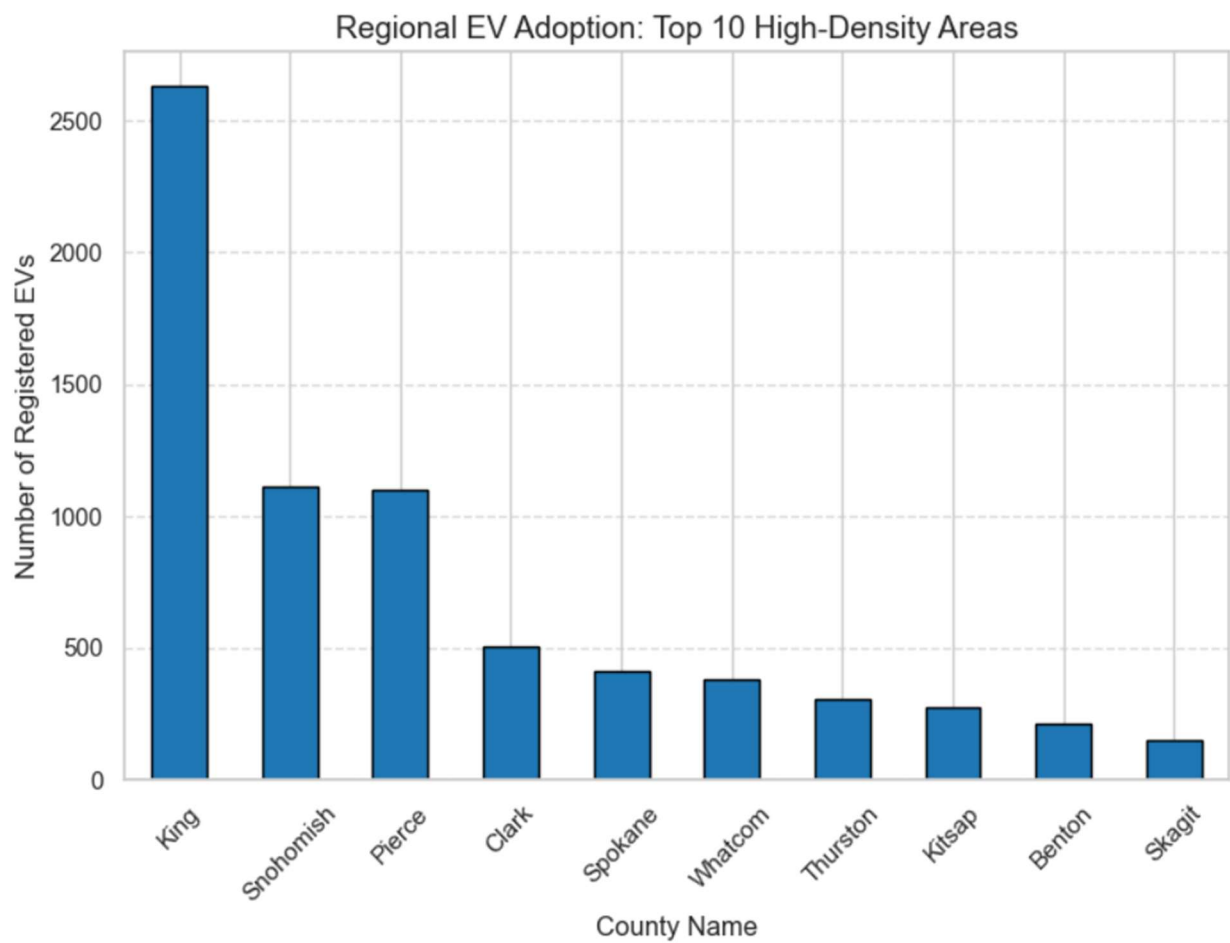
Yes. There is a trend between Urban and Rural areas. The top county accounts for 31.2% of all EVs in this dataset.

TOP 5 COUNTIES (Highly Urbanized Centers)

<u>County</u>	<u>Count</u>
King	2628
Snohomish	1116
Pierce	1104
Clark	503
Spokane	413

BOTTOM 5 COUNTIES (Likely Rural/Low-Density)

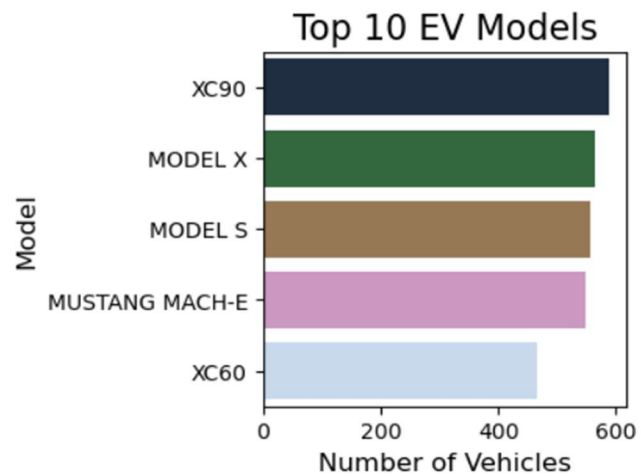
<u>County</u>	<u>Count</u>
Howard	1
Saratoga	1
Richland	1
Bell	1
Virginia Beach	1



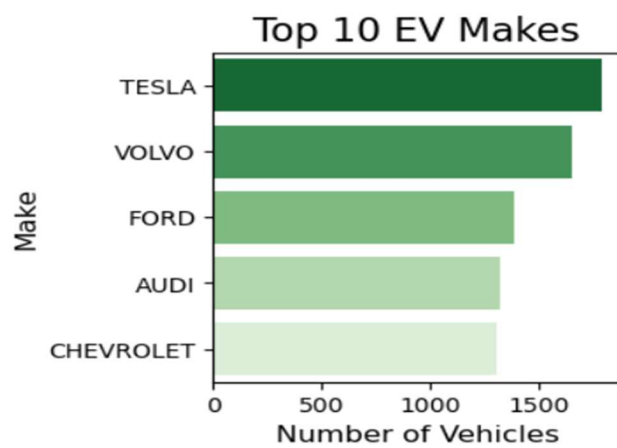
Section 3 - Data Visualization:

1. Bar chart showing the top 5 EV makes and models by count.

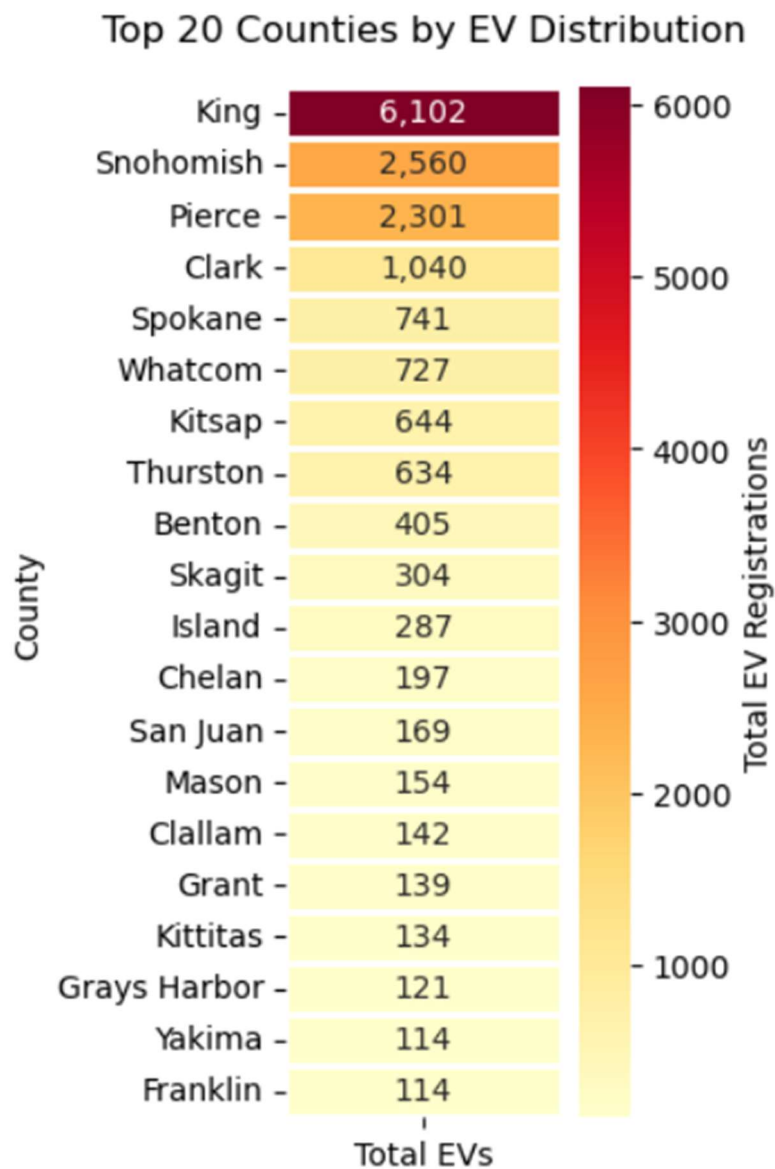
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4	XC60	465



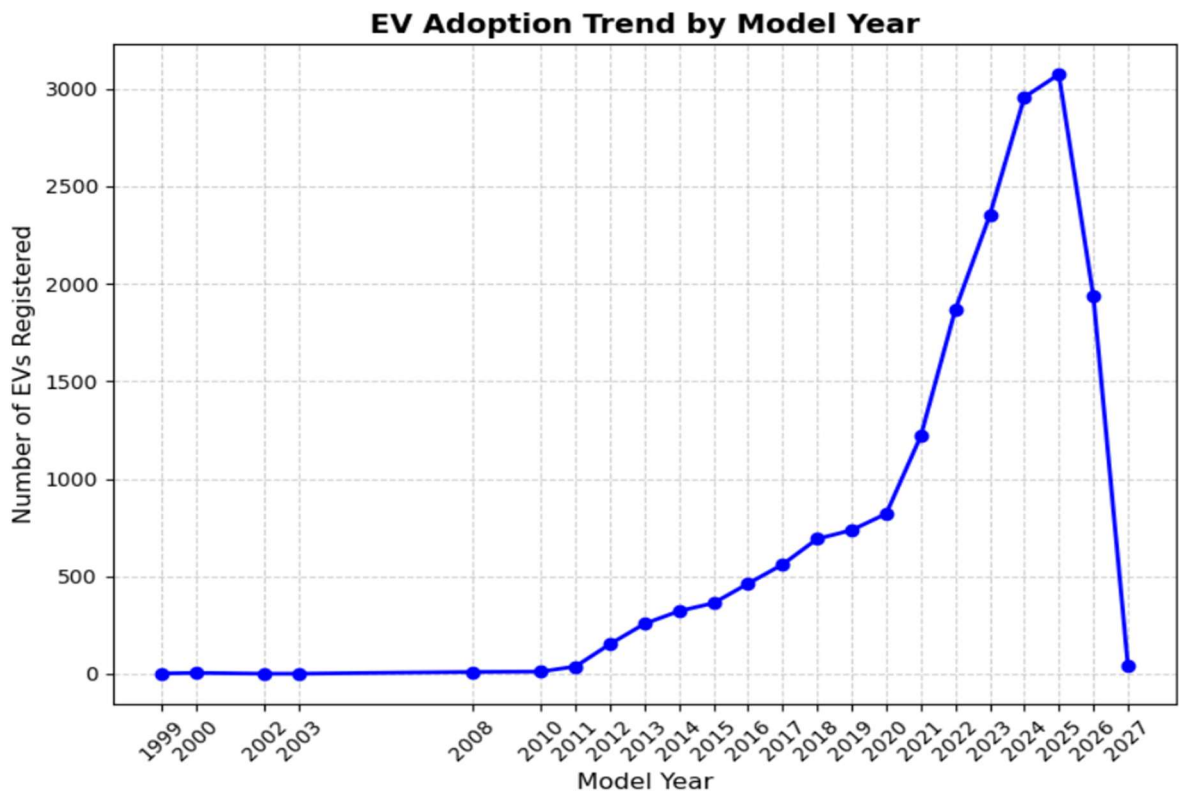
	Make	count
0	TESLA	1787
1	VOLVO	1652
2	FORD	1387
3	AUDI	1317
4	CHEVROLET	1302



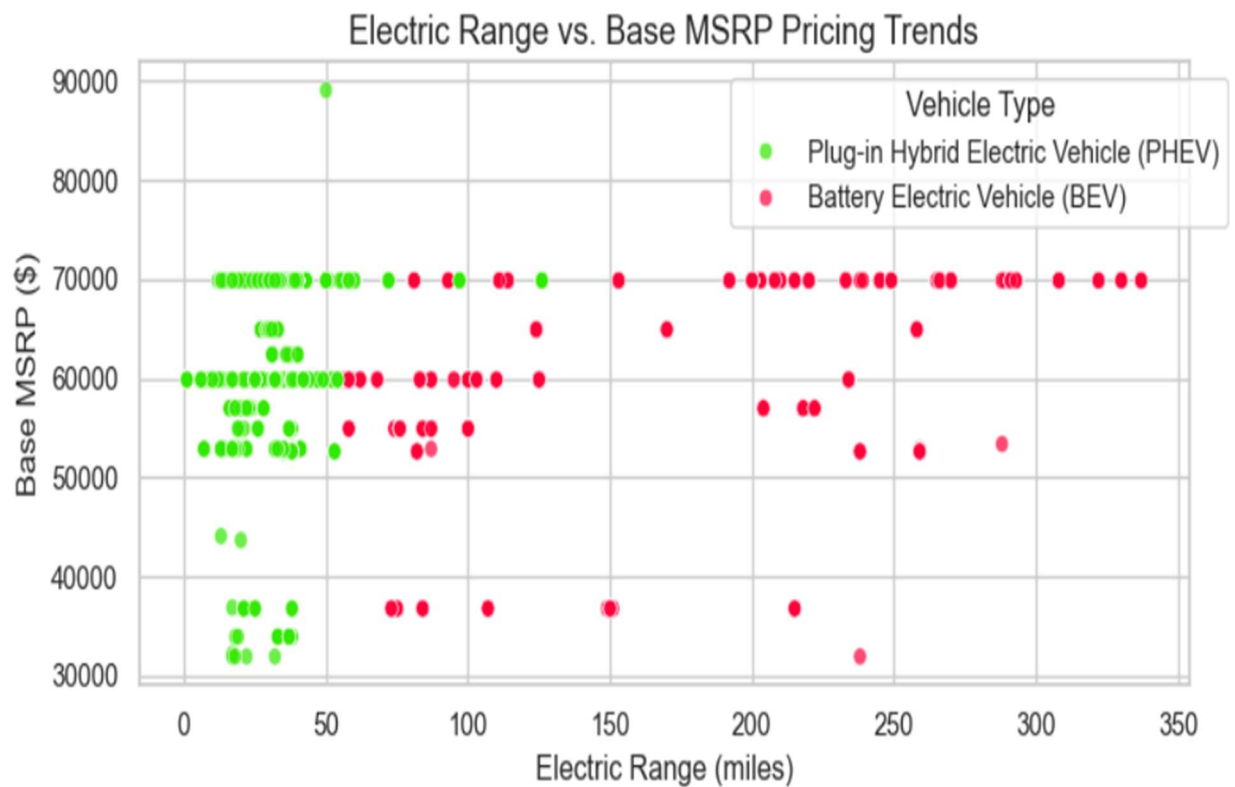
2. Heatmap or choropleth map to visualize EV distribution by county.



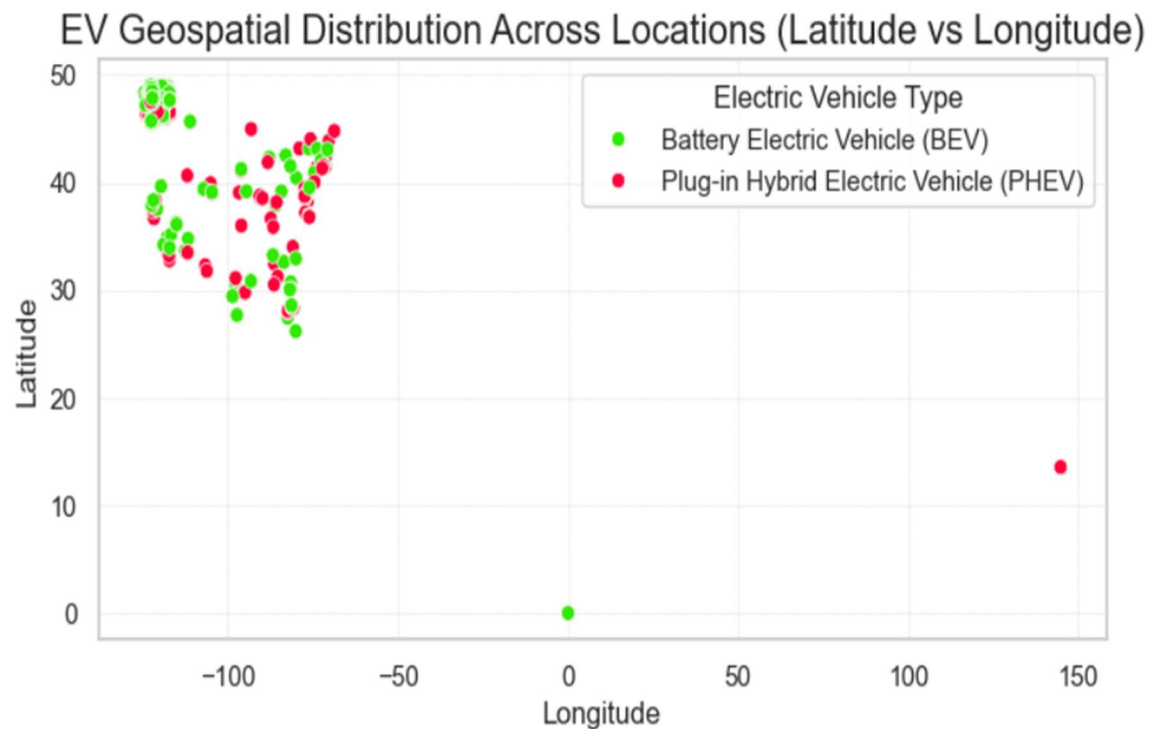
3. Line graph showing the trend of EV adoption by model year.



4. Scatter plot comparing electric range vs. base MSRP to see pricing trends:



6. Use a geospatial map to display EV registrations based on vehicle location.



Section 4 - Linear Regression Model:

1. How can we use Linear Regression to predict the Electric Range of a vehicle?

In this dataset, **Battery capacity metric** is completely missing.

So, to predict the Electric Range of a vehicle using Linear Regression, we can establish a mathematical relationship where **Electric Range is the dependent target (y)**, and the vehicle specifications act as **independent features (X)**

Linear Regression only understands numbers. So, we must transform the dataset's columns

2. What independent variables (features) can be used to predict Electric Range? (e.g., Model Year, Base MSRP, Make)

The columns that hold actual predictive power for Electric Range are:

- **Model Year:** Tracks advancements in battery density over time (Linear feature).
- **Make & Model:** Capture engineering baselines (e.g., Tesla builds different default ranges than Nissan).

- **Electric Vehicle Type:** Tells the model if it is a short-range Plug-in Hybrid (PHEV) or a long-range Battery Electric Vehicle (BEV).

3. How do we handle categorical variables like Make and Model in regression analysis?

- **Make** and **Model** in vehicle datasets) are handled using encoding techniques like **One-Hot Encoding** or **Target Encoding** before running a regression analysis.
- **Make (e.g., Ford, Toyota):** Moderate number of unique categories (low-to-medium cardinality).
- **Model (e.g., Mustang, Camry):** High number of unique categories (high cardinality).
- **The Problem:** Regression models require numerical inputs. Simple label encoding (1, 2, 3...) introduces a fake mathematical order that confuses the model.

Here I have used **One Hot Encoding** method. Because it creates a binary column (0 or 1) for each unique category. It works well for "Make" because there are usually fewer than 50 car manufacturers.

4. What is the R^2 score of the model, and what does it indicate about prediction accuracy?

Model R^2 Score: 0.8793

Mean Absolute Error: 17.50 miles

5. How does the Base MSRP influence the Electric Range according to the regression model?

Base MSRP shows a very weak or negligible positive relationship with electric range, meaning higher vehicle prices do not strongly predict a larger driving range.

6. What steps are needed to improve the accuracy of the Linear Regression model?

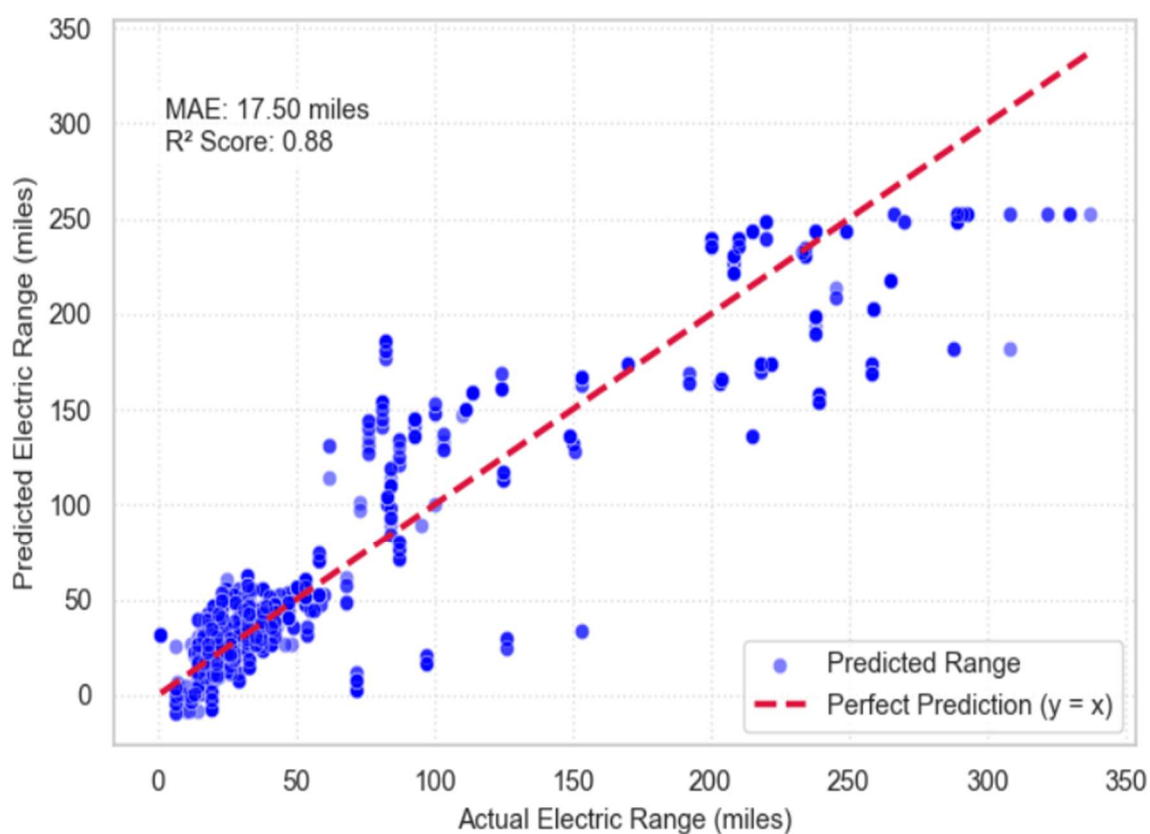
Data cleaning, feature selection, and feature scaling.

1. Drop Invalid Base MSRPs or Filter out missing range entries
2. Check and Remove Outliers
3. Separate features (X)-categorical and target (y)-Numerical
4. Split data into training and testing sets
5. Preprocess categorical data (One-Hot Encoding)
6. Create the Regression Pipeline
7. Train the Model
8. Evaluate Predictions

7. Can we use this model to predict the range of new EV models based on their specifications?

Yes, we can absolutely use machine learning models to predict the range of new Electric Vehicle (EV) models based on their technical specifications.

Linear Regression Evaluation: Actual vs. Predicted EV Range



Conclusion:

- Through exploratory data analysis, the top 5 most common EV makes and models, distribution of EVs by county were determined.
- King County had the most cars registered, followed by Snohomish and Pierce counties.
- A sharp rise in EV was observed I around 2020 to 2025.EV adoption reached peak in the year of 2020.
- Battery Electric vehicle Type is higher than Plug in hybrid vehicle Ev type.
- Count of Clean Alternative Fuel Vehicle Eligible and Non-Eligible Vehicles.
- The top model is VOLVO XC90 car and the Top Make is TESLA overall.
- CAFV eligible vehicles are having higher Electric range.
- The dataset also provided insight into the top 5 EV Makes and Models.
- Geospatial EV Distribution across Locations.
- There is a trend between Urban and Rural areas. The top county accounts for 31.2% of all EVs in this dataset.

